

Economics 2220

Principles Of Macroeconomics

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Lecture 9:

Macroeconomic Relation

► What you will learn in this chapter



- The nature of the multiplier, which shows how initial changes in spending lead to further changes
- The meaning of the aggregate consumption function, which shows how current disposable income affects consumer spending
- How expected future income and aggregate wealth affect consumer spending
- The determinants of investment spending and the distinction between planned investment spending and unplanned inventory investment

THE MULTIPLIER: AN INFORMAL INTRODUCTION



Since households and firms are mutually interdependent (recall the circular flow), booms and busts involve chain reactions.



The multiplier helps us understand the extent of the chain reactions.



Basically, we want to understand how much extra income and spending are created from an initial change in spending.

Example: if UNMC is building a new \$1 billion hospital, how much does this affect the economy?

THE MPC

A lot depends on how much consumers spend when they receive more income

Marginal propensity to consumer, or MPC

$$MPC = \frac{\Delta \text{Consumer spending}}{\Delta \text{Disposable income}}$$

(where Δ = change)

For example, if consumer spending goes up by \$6 and disposable income goes up by \$10, $MPC = \$6/10 = 0.6$

Whatever is not spent is saved, so:

Marginal propensity to save, or MPS = the fraction of an additional dollar of disposable income that is saved.

$$MPS = 1 - MPC$$

THE MULTIPLIER EFFECT

Each \$1 increase in aggregate spending raises both real GDP and disposable income by \$1—and causes people to spend money. **How much?**

If investment spending rises by \$100 billion, this will lead to a second-round increase of $MPC \times \$100$ billion. It is followed by a third-round increase in consumer spending of $MPC \times MPC \times \$100$ billion, and so on up to:

**Total increase in real GDP = $(1 + MPC + MPC^2 + MPC^3 + \dots)$
× \$100 billion OR**

$$= \frac{1}{1-MPC} \times \$100 \text{ billion}$$

LEARN BY DOING: DISCUSS

With a partner, answer the following:

Suppose that the **marginal propensity to save** in an economy is equal to **0.2**. Suppose that the level of **investment spending increases by \$200** this year.

Trace out the total increase in real GDP, showing the **increase in consumer spending** for at least the **second, third, and fourth round of spending** and the **overall increase in real GDP**.

(Assume that in this economy there is no government sector, no taxes, and no transfers and that the aggregate price level and the interest rate are fixed.)

LEARN BY DOING: DISCUSS

(Answer to question on previous slide)

Since $MPS = 0.2$, we know that $MPC = 0.8$.

Increase in investment spending	\$200
+Second-round increase in consumer spending	$= (MPC \times \$200) = \160
+Third-round increase in consumer spending	$= (MPC)(MPC \times \$200) = \128
+Fourth-round increase in consumer spending	$= (MPC)[(MPC)(MPC \times \$200)] = \$102.40$
...Total Increase in real GDP	$= [1/(1 - MPC)] \times (\$200) = \$1,000$

THE MULTIPLIER



If ΔAAS = autonomous change in aggregate spending and

ΔY = change in real GDP

$$\Delta Y = \frac{1}{1-MPC} \times \Delta AAS$$

And the **multiplier** =

$$\frac{\Delta Y}{\Delta AAS} = \frac{1}{1-MPC}$$

LEARN BY DOING: PRACTICE QUESTION

Holding everything else constant in an economy, the larger the *MPS*, the:

- a) smaller the value of the multiplier.**
- b) larger the value of the multiplier.**

SAND STATE SLUMP

Florida, Arizona, California and Nevada were hit harder by the housing bust because the (especially deep) construction slump spread throughout the economy- and incomes fell everywhere.



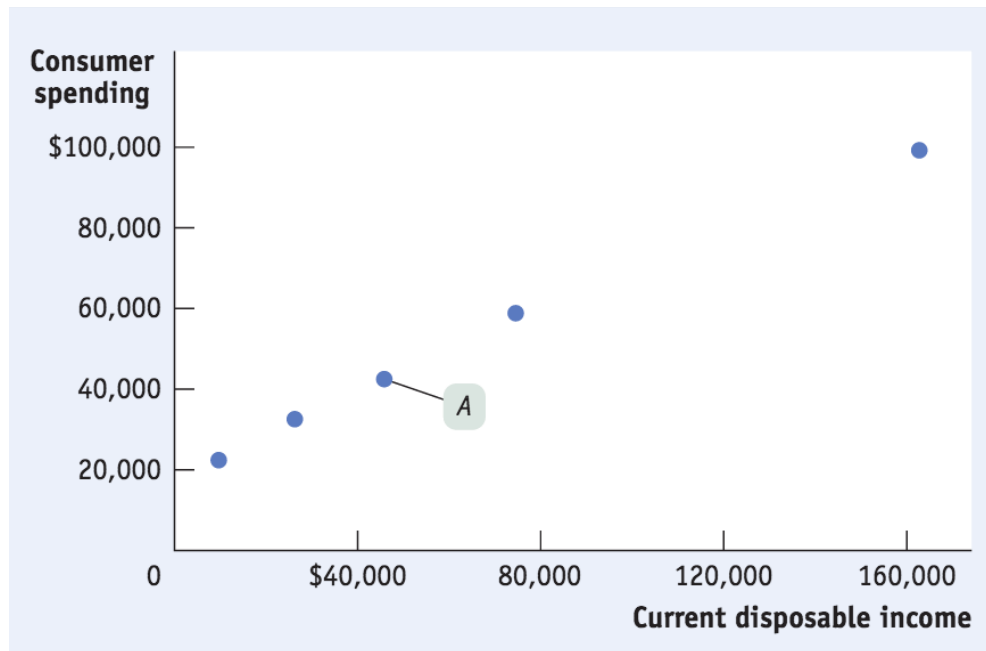
Source: Bureau of Labor Statistics.

CONSUMER SPENDING



Households are constantly confronted with choices—not just about what to consume but also about how much to spend. And **that depends mostly on income.**

CURRENT DISPOSABLE INCOME AND CONSUMER SPENDING



Current disposable income: income after taxes are paid and government transfers are received. **2013: average household disposable income = \$45,826 and average household consumer spending = \$42,495**

THE CONSUMPTION FUNCTION

Consumption function: an equation showing how an individual household's consumer spending varies with the household's disposable income.

$$c = a + MPC \times yd$$

Where

c = a household's consumer spending

yd = household disposable income

MPC = marginal propensity to consume

a = a constant, autonomous consumer spending—what a family would spend even with zero income

THE CONSUMPTION FUNCTION

Recall,

$$MPC = \Delta c / \Delta yd$$

Multiplying both sides of the equation by Δyd , we get:

$$MPC \times \Delta yd = \Delta c$$

In other words, when y_d goes up by \$1, c goes up by $MPC \times \$1$.

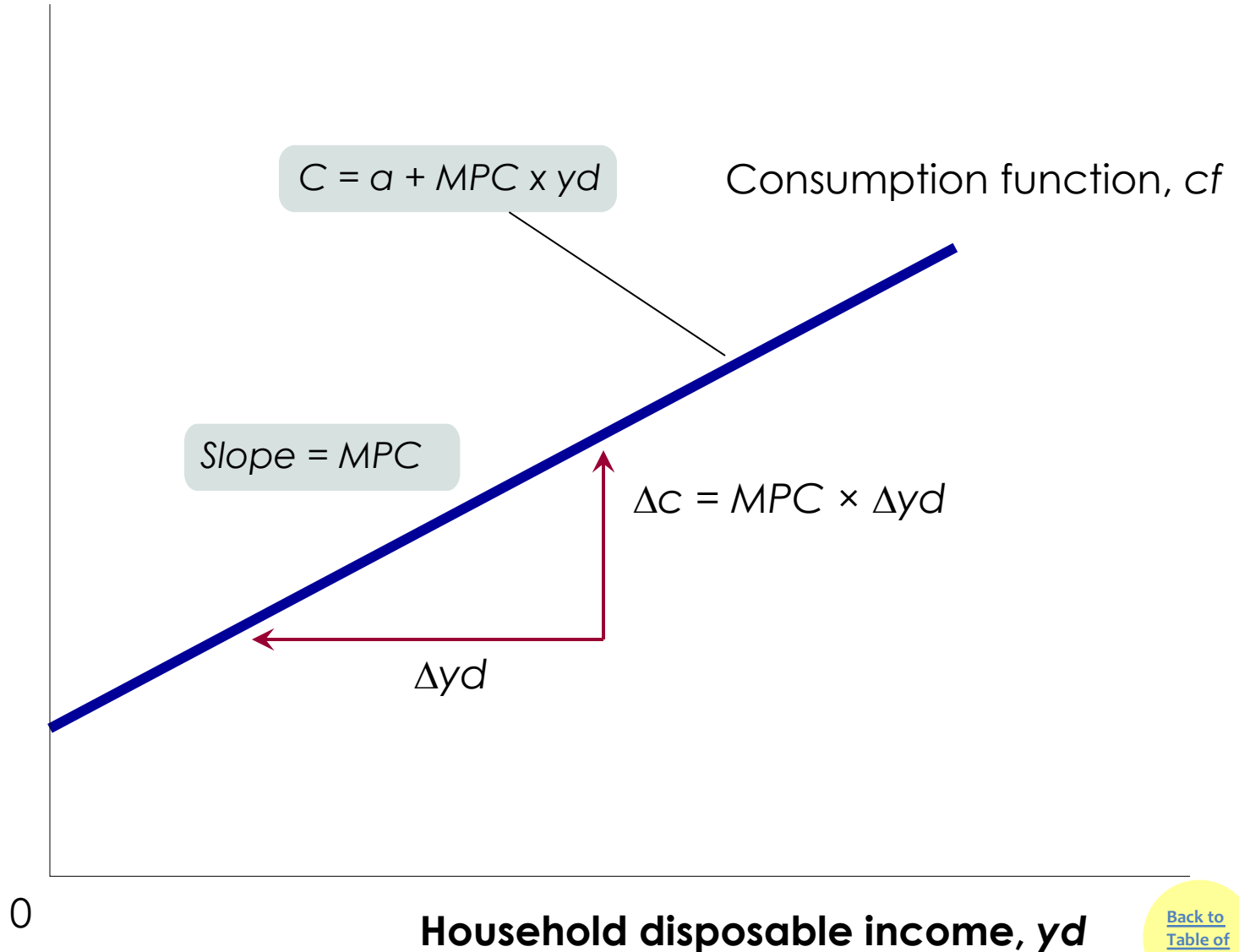
LEARN BY DOING: PRACTICE QUESTION

Suppose when Sue's disposable income is \$10,000, she spends \$8,000, and when her disposable income is \$20,000, she spends \$14,000. Sue's autonomous consumer spending is equal to _____ and her *MPS* is equal to _____.

- a) \$0; 0.2
- b) \$2,000; 0.2
- c) \$0; 0.4
- d) \$2,000; 0.4

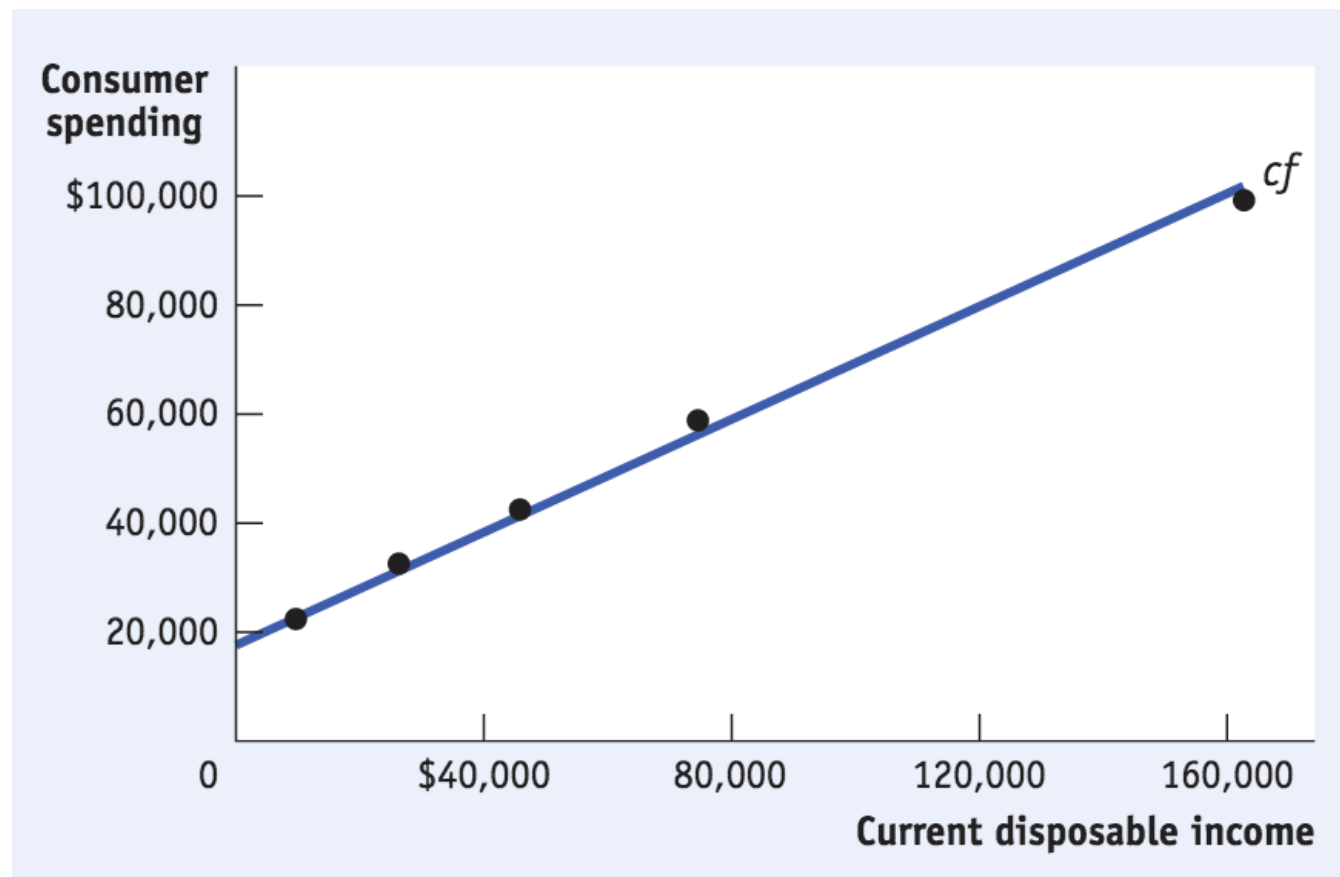
THE CONSUMPTION FUNCTION

Household
consumer
spending, c



HOW WELL DO THE DATA FIT THE THEORY?

Pretty well. Here's 2013. Autonomous spending is around \$19,343 and $MPC = 0.50$.



AGGREGATE CONSUMPTION FUNCTION

Aggregate consumption function: the relationship for the economy as a whole between aggregate disposable income and aggregate consumer spending.

$$C = A + MPC \times YD$$

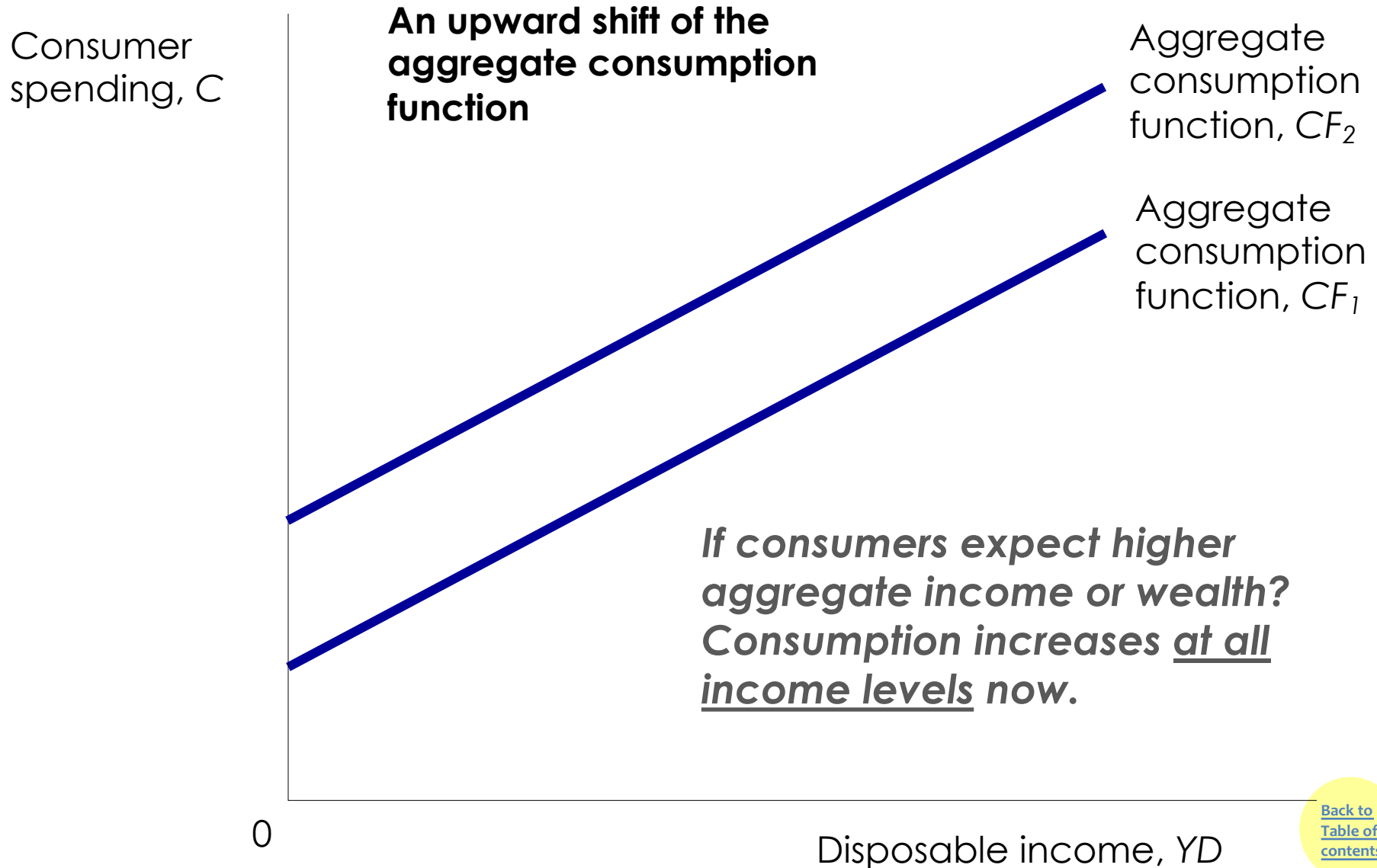
Same form as consumption function, just aggregate.

LEARN BY DOING: PRACTICE QUESTION

Assume aggregate consumer spending equals \$5,000 when aggregate disposable income is zero, and when disposable income increases from \$300 to \$400, consumer spending increases by \$70. What is the equation for the aggregate consumption function?

- a) $C = 5,000 + 70YD$
- b) $C = 500 + 0.7YD$
- c) $C = 5,000 + 0.7YD$
- d) $C = 5,000 + 7YD$

SHIFTS IN THE AGGREGATE CONSUMPTION FUNCTION



SHIFTS IN THE AGGREGATE CONSUMPTION FUNCTION

Consumer
spending, C

A Downward shift of the aggregate consumption function

*If consumers expect lower
aggregate income or wealth?
Consumption decreases at all
income levels now.*

Aggregate
consumption
function, CF_1

Aggregate
consumption
function, CF_2

0

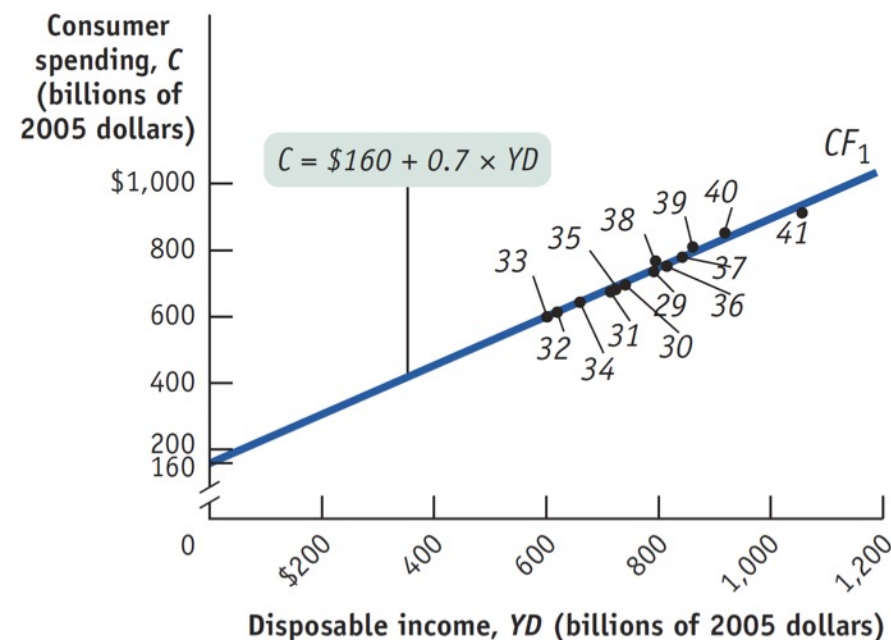
Disposable income, YD

SHIFTS OF THE AGGREGATE CONSUMPTION FUNCTION

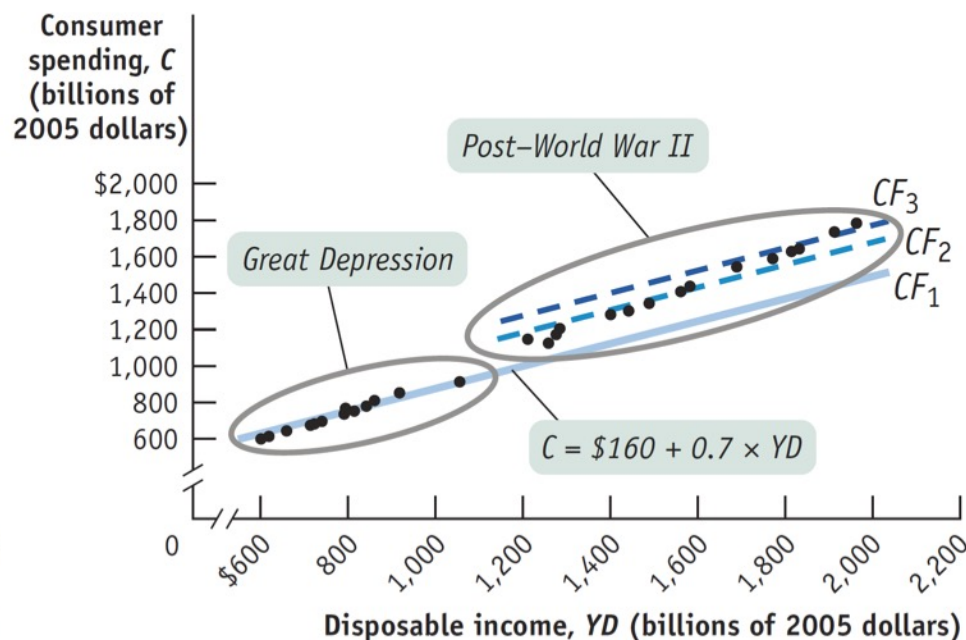
What causes shifts?

changes in expected future disposable income
changes in aggregate wealth

(a) A Simple Aggregate Consumption Function Fits Data from the Great Depression Years Quite Well . . .



(b) . . . but Greatly Underestimates Consumer Spending After the War.

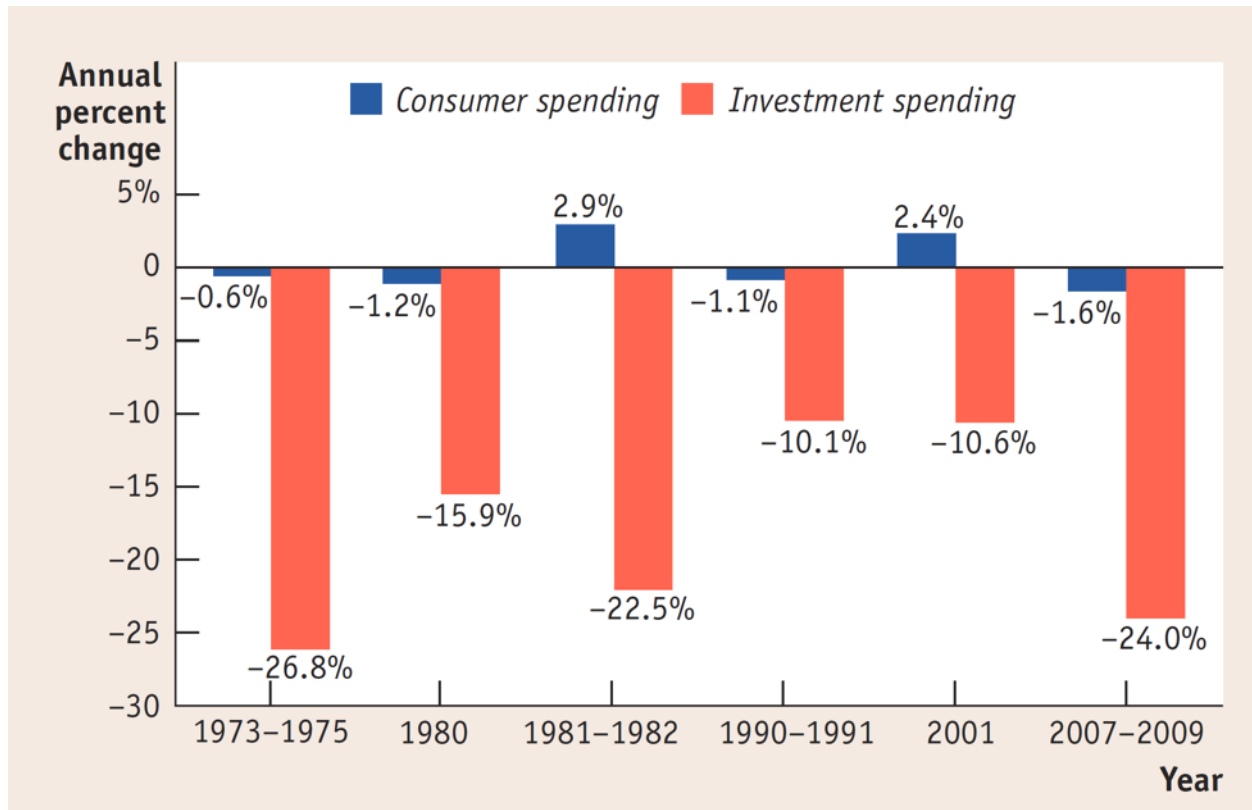


Did the aggregate consumption function shift after WWII? It looks that way.

Source: Bureau of Economic Analysis

INVESTMENT SPENDING

Small yet powerful: Although much smaller than consumer spending, **investment spending tends to drive the booms and busts in the business cycle.** Shown here: 6 recent recessions



WHAT DRIVES (PLANNED) INVESTMENT SPENDING?

Planned investment spending: the investment spending that businesses intend to undertake during a given period.

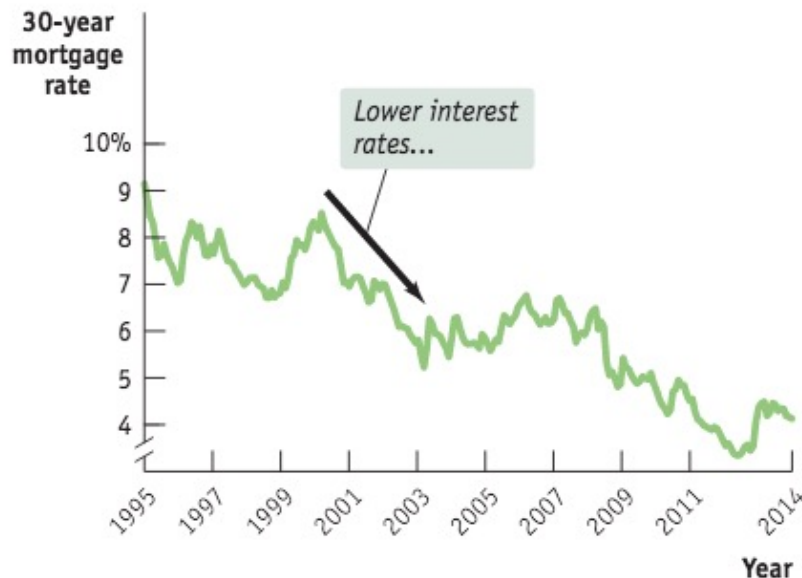
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- A hand holding a blue pen points to a clipboard held by another hand. The clipboard has a white sheet of paper with a list of three factors in blue text. The person holding the pen is wearing a blue long-sleeved shirt.
1. The interest rate
 2. Expected future real GDP
 3. Current level of production capacity

THE INTEREST RATE AND INVESTMENT SPENDING

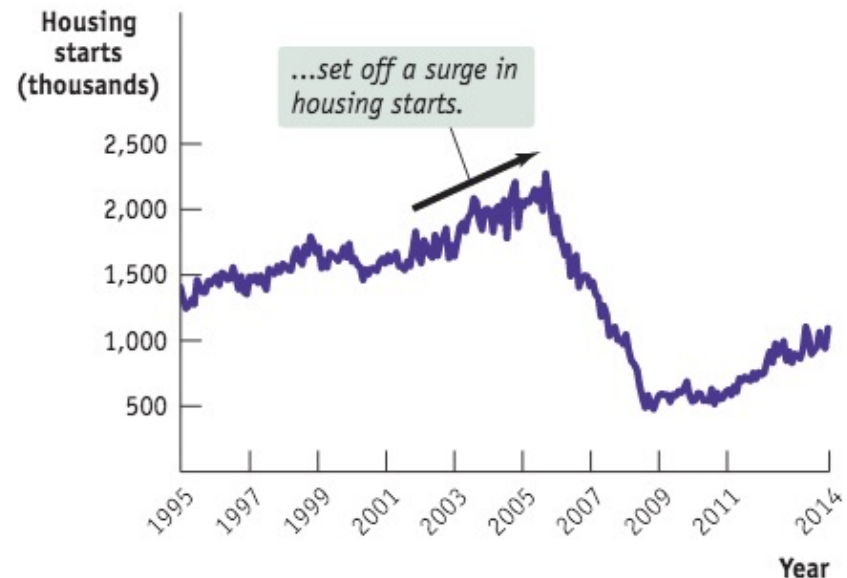
Interest rates are often the cost of investment projects.

When interest rates are low, more loans are undertaken and investment rises (other things equal).

(a) The Interest Rate on 30-Year Mortgages



(b) Housing Starts



Source: Federal Reserve Bank of St. Louis.

EXPECTED FUTURE REAL GDP, PRODUCTION CAPACITY, AND INVESTMENT SPENDING

If your firm has extra capacity and doesn't expect sales to increase, its investment will be lower.

According to the *accelerator principle*:

A higher rate of growth in real GDP leads to higher planned investment spending.

A lower growth rate of real GDP leads to lower planned investment spending.

INVENTORIES AND UNPLANNED INVESTMENT SPENDING

Inventories: stocks of goods held to satisfy future sales.

Inventory investment: the value of the change in total inventories held in the economy during a given period.

Unplanned inventory investment: unplanned changes in inventories occurring when actual sales are more or less than businesses expected.

Actual investment spending: the sum of planned investment spending and unplanned inventory investment.

So in any period: $I = I_{\text{Unplanned}} + I_{\text{Planned}}$